

one of the several uses of gears. Gears are used when one needs to control the speed or torque of a moving element.

A gear is a rotating machine part which has teeth-like cuts on one of its rotatable sides. When two or more of these parts work together (see picture), the system is called a transmission. Gears are analogous to pulleys, however they



have an added advantage – gears prevent slipping because of their teeth-cut design. The most popular and commonly seen example of the gear is the Spur gear used to adjust rotational speeds. Other types of gears transform the direction of motion as well. An interesting ‘Sun and Planet’ gear system was used by Scottish inventor, James Watt on his early engines instead of a crank system for power transmission.

Most modern day transport vehicles employ gears to make the best use of the power they generate by burning fuel. Without gears, journeys would have been slower, inefficient and quite costly. Interestingly, compact engines inside high-speed rides ranging from bikes to planes are benefitted heavily by gears, without which they would be huge, unreliable, inefficient, and once again, costly.

Geiger Counter

Geiger Counter is an instrument that detects and measures the intensity of radiation. It's a metal cylinder filled with low-pressure gas sealed by a plastic or ceramic window at one end. A thin metal wire made of tungsten runs down the centre of the tube. This wire is connected to a high, positive voltage resulting in a strong electric field between it and the outside tube.

When radiation enters the tube, it causes ionization, splitting gas molecules into ions and electrons. The negatively charged electrons get pulled by the high-voltage positive wire and as they zoom through the tube collide with more gas molecules and produce further ionization. The result